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Roll No

IT-703(D)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Digital Image Processing

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss the fundamental steps involved in digital image processing.
b) What are the elements of digital image processing?
2. a) List out different noise probability density functions used in image processing applications?
b) With necessary equations, explain about Homomorphic filtering.
3. a) Discuss how the various filter masks are generated to sharpen images in spatial filters.
b) Explain spatial filtering in image enhancement.
4. a) Draw and explain the general image compression system model.
b) Explain the significance of thresholding in image segmentation.

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5. a) Compute the Haar transform of the 2×2 image

$$F = \begin{bmatrix} 3 & -1 \\ 6 & 2 \end{bmatrix}$$

- b) Explain about image restoration using minimum mean square filtering.
6. a) Explain image sharpening using Butterworth high pass and Gaussian high pass filters?
- b) Discuss about region based segmentation.
7. a) What is Morphology? Discuss basic morphological Algorithms.
- b) Explain boundary and regional descriptors?
8. Write short notes
- a) Pattern and Pattern Classes
 - b) Geometric Mean Filter
 - c) Edge linking
 - d) Sampling
